

EARLY DETECTION OF RED PALM WEEVIL (RPW) USING EMBEDDED ACOUSTIC SIGNAL PROCESSING

SCSBDPROJ - DESIGN THINKING AND INNOVATIONS

Submitted in partial fulfilment of the requirements for the award of
Bachelor of Engineering degree in Electronics and Communication Engineering

by

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**DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING
SCHOOL OF ELECTRICAL AND ELECTRONICS**

SATHYABAMA

**INSTITUTE OF SCIENCE AND TECHNOLOGY
(DEEMED TO BE UNIVERSITY)**

Accredited with Grade "A++" by NAAC

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BONAFIDE CERTIFICATE

This is to certify that this Project Report is the Bonafide work of **K. Siva Srinivas** (44130218), **K. Keshav** (44130225), **K. Lokesh** (44130244) , **S. Kishore Vikhram** (44130229) who carried out the project entitled "**EARLY DETECTION OF RED PALM WEEVIL (RPW) USING EMBEDDED ACOUSTIC SIGNAL PROCESSING**" under our supervision from November 2025 to April 2026.

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We, **K. Siva Srinivas** (44130218), **K. Keshav** (44130225), **K. Lokesh** (44130244), **S. Kishore Vikhram** (44130229), hereby declare that the Project Report entitled **"Early Detection of Red Palm Weevil (RPW) using Embedded Acoustic Signal Processing"** done by me under the guidance of **Dr. G. RAJALAKSHMI, M.E., Ph.D. Associate Professor** is submitted in partial fulfillment of the requirements for the award of Bachelor of Engineering degree in Department of Electronics and Communication Engineering.

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ABSTRACT

Red Palm Weevil (RPW), *Rhynchophorus ferrugineus*, is the most destructive pest of palm trees worldwide and is responsible for severe economic losses in coconut farming across Tamil Nadu and Kerala. The larvae feed deep inside the trunk for weeks before any external symptom appears, and by the time crown drooping is visible, internal tissue destruction has typically reached 70 to 80 percent. Detection must therefore happen before visible symptoms — while the larvae are still actively chewing. Existing approaches either detect the pest too late (visual inspection), miss the larval phase entirely (pheromone traps), or require expensive equipment and cloud connectivity that is impractical for small Indian farms.

This project presents a fully offline, low-cost embedded acoustic detection system built around an ESP32-WROOM-32 microcontroller and a piezoelectric contact sensor. The system captures structure-borne vibrations from the palm trunk through a waveguide screw and processes the signal entirely on-device using the Teager-Kaiser Energy Operator (TKEO). Unlike frequency-domain FFT analysis, TKEO detects the temporal burst pattern of larval chewing — a characteristic that reliably separates infestation from environmental noise such as wind, rain, and machinery. The detection pipeline runs at 8 kSPS with a computational cost of just three multiplications per sample, making real-time processing on the ESP32 trivial.

The system produces a three-state output: Normal, Suspicious, and Infested — with a persistence escalation rule that reduces false alarms. Laboratory evaluation on simulated chewing signals achieved 94% sensitivity and a 6.7% false positive rate at a wind noise level of 65 dB SPL. Total hardware cost per unit is INR 910, battery life is approximately 15 days per charge, and the system requires no internet connectivity.

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CHAPTER 1

INTRODUCTION

1.1 BACKGROUND AND MOTIVATION

In the 21st century, with the rapid expansion of agricultural activity and the growing threat of invasive pests, traditional crop protection systems are becoming increasingly inadequate. Nowhere is this clearer than in the coconut farming regions of Tamil Nadu and Kerala, where Red Palm Weevil (RPW), *Rhynchophorus ferrugineus*, has emerged as the single most damaging pest facing smallholder farmers. India loses an estimated 20 to 25 percent of its annual coconut yield to RPW infestation annually, and the true figure is likely higher because many attacks go undetected until the tree is already unsalvageable.

The biology of the pest is the root cause of the detection problem. Adult females lay eggs beneath leaf axils or in bark wounds, and the hatched larvae spend their entire destructive phase feeding deep inside the trunk, boring tunnels through the fibrovascular tissue. The tree shows no external symptoms during this period. Crown drooping, yellowing fronds, and the characteristic fermented odor of decaying tissue only appear when internal damage has already reached 70 to 80 percent. At that stage, curative treatment is not possible. The only practical response is felling the tree to prevent adult weevils from spreading to neighboring palms — a response that is costly, permanent, and demoralizing for a small farmer.

Traditional detection methods fail at exactly this stage. Visual inspection misses the critical window entirely, because there is nothing to see. Pheromone traps catch adult weevils but provide no information about whether larvae are already active inside the trunk. Laboratory-grade acoustic systems can detect larval chewing, but they require expensive hardware and skilled operators. Machine learning approaches achieve good classification accuracy on curated datasets, but demand cloud connectivity and high-power compute — neither of which is realistic for a remote coconut farm.

Coconut farming forms the backbone of the rural economy across large parts of coastal Tamil Nadu and Kerala. Millions of smallholder households depend on it as a primary or supplementary income source. The RPW threat has been officially recognized as a major agricultural emergency in both states, with government advisories recommending early detection as the primary control strategy. However, the detection tools currently available to farmers are either too expensive (commercial acoustic systems at ₹15,000 or more per tree), too late (visual inspection), or entirely impractical for rural deployment (cloud-connected ML systems).

This project was motivated directly by this gap. The question we started from was: what would it actually take to give a coconut farmer in a remote coastal village a tool that detects RPW early, is cheap enough to deploy at scale, and does not require a smartphone, internet connection, or technical training? The answer shaped every design decision in this system.

1.2 PROBLEM STATEMENT

The key technical challenge is distinguishing RPW larval chewing from background noise. Larval chewing produces structure-borne mechanical vibrations in the trunk, but so do wind, rain, footsteps, and farm machinery. Prior acoustic detection systems attempted to separate these using frequency-domain features, but the spectral content of RPW chewing overlaps substantially with these noise sources, leading to unacceptably high false positive rates in real field conditions.

The insight that drives this project is biological: RPW larvae do not chew continuously. Chewing happens in short impulse bursts separated by rest periods. This temporal structure — short spikes grouped into bursts, separated by gaps — is fundamentally different from all common environmental noise sources, which produce either continuous broadband vibration (wind, rain) or periodic signals (machinery). A detector sensitive to burst temporal structure rather than spectral content will naturally reject these noise sources. The Teager-Kaiser Energy Operator (TKEO) is ideally matched to this task: it estimates instantaneous energy with single-sample time resolution, making it far better suited to detecting short chewing spikes than a multi-millisecond FFT frame.

CHAPTER 2

LITERATURE SURVEY

2.1 REVIEW ON LITERATURE SURVEY

Mankin et al. (2011) — This foundational study demonstrated that piezoelectric contact sensors can detect internal larval activity in tree trunks by picking up structure-borne vibrations. The authors used frequency-domain features and showed clear separation between larval acoustic signatures and background noise in controlled laboratory conditions. However, the system was never designed for embedded deployment, and field trials showed significant false positive rates when wind and machinery noise was present — the spectral content overlaps considerably with larval chewing in the 1–3 kHz band. This paper established the viability of acoustic RPW detection and remains the most cited work in the field.

Reference: Mankin, R. W., Brandhorst-Hubbard, J., Flanders, K. L., Zhang, M., and Crouch, R. (2011). Acoustic detection of the red palm weevil. *Journal of Economic Entomology*, 104(5), 1471–1481.

Pinhas et al. (2008) — This work made the key biological observation that motivates the present project: RPW larval chewing is impulsive and intermittent, producing bursts of short spikes separated by rest periods. The authors noted that this temporal structure differs fundamentally from environmental noise sources. They used this observation for signal segmentation rather than as the primary detection feature, but it pointed clearly toward a temporal energy approach. Their detection accuracy in laboratory conditions reached approximately 83%.

Reference: Pinhas, J., Soroker, V., Hetzroni, A., Mizrach, A., Teicher, M., and Goldenberg, J. (2008). Automatic acoustic detection of the red palm weevil. *Computers and Electronics in Agriculture*, 63(2), 131–139.

Potamitis (2020) — This study applied convolutional neural networks to insect acoustic datasets, including RPW, and demonstrated high classification accuracy above 92% on curated test data. The system required a server for training and inference, with no embedded deployment path. The dataset was collected in quiet laboratory conditions, and the authors acknowledged that generalisation to outdoor noise environments was an open problem.

Reference: Potamitis, I. (2020). Automatic recording and identification of insects with remote sensing methods. *Sensors*, 20(19), 5625.

Hetzroni et al. (2016) — This research used Support Vector Machines on spectral features extracted from contact microphone recordings of RPW-infested date palms. Field trials were conducted, and the system achieved approximately 88% accuracy. Processing was performed on a laptop connected to the sensor — there was no embedded component. The work demonstrated that field validation is significantly harder than laboratory characterisation, particularly in the presence of wind noise.

Reference: Hetzroni, A., Mizrach, A., Teicher, M., and Timar-Gabor, A. (2016). Machine learning approach for acoustic detection of the red palm weevil. *Journal of Pest Science*, 89(2), 539–549.

Kaiser (1990) — The original paper introducing the Teager-Kaiser Energy Operator, defined as $\psi[x(n)] = x^2(n) - x(n+1) \cdot x(n-1)$. Kaiser showed that this three-sample operator provides an instantaneous energy estimate with far better transient localization than squared-amplitude or Hilbert-transform approaches. The TKEO has since been widely applied in speech processing and biomedical signal analysis for detecting short-duration energy events in noisy signals — properties that make it directly applicable to RPW burst detection.

Reference: Kaiser, J. F. (1990). On a simple algorithm to calculate the "energy" of a signal. In *Proceedings of IEEE ICASSP*, vol. 1, pp. 381–384.

Sutin et al. (2021) — This study explored active ultrasonic sensing combined with passive acoustic listening for RPW detection. The active approach allowed the system to probe the structural integrity of the trunk and cross-reference with passive acoustic signatures. While the results were promising in laboratory settings, the equipment cost was substantially higher than passive-only approaches, and field deployment was not addressed.

Reference: Sutin, A., Liberman, B., and Salloum, H. (2021). Acoustic detection of red palm weevil in palm trees using active ultrasound and passive listening. *Acoustics*, 3(1), 1–14.

Faleiro (2006) — A comprehensive review of RPW biology, infestation patterns, and management strategies in Asia and the Middle East. This paper provides essential context for understanding why early detection is critical: by the time the pest is visible, the tree is already dead. The review covers pheromone-based monitoring, quarantine measures, and chemical treatment, noting that none of these approaches addresses the detection window between egg-laying and visible symptom onset.

Reference: Faleiro, J. R. (2006). A review of the issues and management of the red palm weevil *Rhynchophorus ferrugineus* in Asia and Middle East. *Pesticide Research Journal*, vol. 18, no. 1, pp. 1–24.

Abo-Zahhad et al. (2021) — This paper proposed an IoT-based sensor network architecture for continuous agricultural monitoring using cloud connectivity. While technically sound, the system's dependence on cloud infrastructure and persistent connectivity makes it unsuitable for the rural Indian deployment context targeted in this project, where network coverage is patchy and operational costs must be near zero.

Reference: Abo-Zahhad, M., Ahmed, S. M., and Abd-Allah, M. K. (2021). IoT-based smart agriculture monitoring system. *Computers and Electronics in Agriculture*, 182, 106017.

2.2 INFERENCES AND CHALLENGES IN EXISTING SYSTEM

2.2.1 Inferences

Acoustic detection is biologically justified: The literature consistently confirms that RPW larvae produce detectable acoustic signatures during their feeding phase — well before any visible symptom. This means early detection is acoustically possible; the question is how to do it reliably and cheaply.

Temporal structure is more discriminative than spectral content: The most practically useful observation in the literature (Pinhas et al., 2008) is that RPW chewing is impulsive and intermittent, structurally different from continuous environmental noise. This suggests that a temporal energy approach like TKEO will outperform frequency-domain methods in real field conditions — a prediction confirmed by this project.

Embedded deployment has not been attempted: None of the reviewed works deployed the detection algorithm on a microcontroller or battery-powered embedded platform. All systems used laptop-grade or server-grade compute. This is the primary gap this project addresses.

Field validation remains the hard problem: Most systems achieve good laboratory results but degrade significantly in field conditions due to environmental noise. The adaptive thresholding approach in this project is designed to address this by tracking local noise statistics rather than using fixed thresholds.

2.2.2 Challenges

Frequency overlap with noise: The spectral content of RPW chewing overlaps with wind, rain, and machinery in the 500 Hz to 3 kHz band. Frequency-domain classifiers trained in quiet conditions fail in the field because they cannot distinguish RPW signals from common noise sources spectrally.

Compute constraint: Deploying inference on a microcontroller limits the algorithm complexity that is practical. FFT computation requires hundreds of multiplications per frame; neural network inference is entirely out of reach without dedicated hardware. The TKEO requires three multiplications per sample, fitting comfortably within the ESP32's compute budget.

Cost constraint: Commercial acoustic systems cost ₹15,000 or more per tree. For a 100-tree coconut farm this is ₹15 lakh — completely out of reach for a smallholder. The system designed in this project achieves hardware cost of ₹910 per unit\\.

Dataset scarcity: Real RPW acoustic data from live infested trees is difficult to collect and label. The larval feeding phase must be caught at the right instar stage, under controlled conditions, with verified ground truth. This is why machine learning approaches trained on lab datasets fail to generalise: the training distribution does not represent real-world variability in larval stage, tree species, moisture content, and ambient noise.

CHAPTER 3

EXISTING SYSTEM

3.1 DESCRIPTION OF EXISTING SYSTEMS

3.1.1. PHEROMONE TRAP SYSTEMS:

Pheromone traps are the most widely deployed RPW monitoring tool in India and across the Middle East. They work by releasing synthetic aggregation pheromones (principally ferrugineol) that attract adult male and female weevils to a bucket trap. The trap captures adults and allows farm operators or extension officers to estimate the weevil population pressure in a given area.

The fundamental limitation of pheromone traps is that they detect adult weevils — not larvae. By the time adults are flying and responding to pheromone lures, eggs may already have been laid weeks earlier. Larvae could be feeding inside multiple trees while the trap catches their parents. The trap provides no information about whether any specific tree is currently infested. It is a population indicator, not a tree-level early detection tool.



Fig:3.1: Conventional Pheromone Trap [Source: Agricultural extension documentation]

3.1.2. VISUAL INSPECTION:

Visual inspection by agricultural officers or trained farmers is the most common tree-level detection method. Inspectors look for external symptoms including crown drooping, yellowing or wilting of inner fronds, brownish fluid oozing from the trunk, irregular holes in the soft tissue at the base of leaf petioles, and the fermented odour of decomposing tissue. In advanced cases, the sound of larval feeding can sometimes be heard by pressing an ear against the trunk.

The critical failure of visual inspection is that all these symptoms appear only after 70 to 80 percent of internal tissue has been destroyed. The tree is structurally dead before it looks sick. Treatment at this stage consists only of felling and burning the palm to prevent adult emergence and spread to neighbouring trees. There is no curative option. Visual inspection catches infestations that are already catastrophic and misses entirely the window during which intervention would be effective.



Fig:3.2: Manual Inspection of Palm Trunk showing Crown Drooping [Source: Tamil Nadu Agricultural University]

3.1.3. LABORATORY-GRADE ACOUSTIC DETECTION:

Research-grade acoustic detection systems use piezoelectric sensors or sensitive microphones coupled to the tree trunk. The sensor signal is processed using frequency-domain analysis (FFT) or spectral feature extraction, and the resulting features are classified using trained models (SVM, CNN) running on a connected computer. These systems have demonstrated detection sensitivities of 80 to 90 percent in controlled laboratory conditions.

The reasons these systems have not achieved field deployment are straightforward. First, the equipment cost is high — a research-grade piezoelectric accelerometer alone can cost thousands of rupees, before adding the data acquisition system and compute hardware. Second, processing requires a laptop or embedded computer with substantial battery capacity, making battery life too short for unattended deployment. Third, the frequency-domain classification approach is highly sensitive to ambient noise conditions: models trained in quiet laboratory environments fail to generalise to noisy outdoor farms. Fourth, the systems require a trained operator to set up, calibrate, and interpret results.



Fig:3.3: Lab-Grade Acoustic Detection Setup [Source: Mankin et al., 2011]

3.1.4. DAMAGE PROGRESSION AND DETECTION WINDOW:

Understanding the timeline of RPW infestation is critical to appreciating why early detection matters. After egg-laying, the first instar larvae begin feeding within a few

days. For the first three to four weeks, the tree shows no external symptoms whatsoever. Larval feeding produces audible and measurable acoustic vibrations throughout this phase. Between four and eight weeks after egg-laying, internal damage begins to compromise structural integrity, but surface symptoms are still absent or minimal. Visible crown symptoms appear only after eight to twelve weeks, by which point the tree is beyond saving.

The acoustic detection window — the period during which larval feeding produces detectable signals but the tree is still saveable — spans approximately weeks two through eight after egg-laying. This project targets detection within this window.



Fig:3.4: RPW Damage Progression Stages showing acoustic detection window (weeks 2–8)

3.2 AIM

The aim of this project is to develop a low-cost, battery-powered, fully offline embedded acoustic detection system capable of identifying Red Palm Weevil larval infestation in palm trees before any visible external symptoms appear. The system targets coconut farmers in Tamil Nadu and Kerala who need a practical, affordable detection tool that can be operated without technical training or network connectivity.

The detection is achieved by capturing structure-borne vibrations from the palm trunk using a piezoelectric contact sensor coupled through a waveguide screw,

digitising the signal with an ESP32 microcontroller, and applying the Teager-Kaiser Energy Operator to detect the temporal burst pattern of larval chewing. The system requires no frequency matching, no machine learning inference, and no cloud server.

3.3 SCOPE

The scope of this project covers the complete design, hardware implementation, firmware development, and laboratory characterisation of the RPW acoustic detection system. This includes sensor coupling design, analog front-end circuitry, ESP32 firmware in embedded C++, the TKEO-based DSP pipeline, adaptive thresholding, burst detection and decision logic, power management, and mechanical packaging.

The scope extends to two deployment modes: a portable probe for periodic tree-by-tree scanning surveys, and a permanent per-tree installation for continuous background monitoring. Both modes are designed for operation by non-specialist users with no technical background beyond a brief demonstration.

Future extensions within scope but not implemented in this version include LoRa radio integration for multi-tree network coverage, solar charging for perpetual operation, and a small binary Tiny-ML classifier for improved burst discrimination. Field validation on live RPW-infested trees is identified as the immediate next step beyond the scope of this report.

Identified Hardware and Software Improvements for Next Phase:

The following hardware and software improvements have been identified through critical design review and are planned for implementation in the next revision of the system.

(i) Op-Amp Replacement (Hardware): The LM358 op-amp used in the current analog front end has a gain-bandwidth product of 1 MHz and a slew rate of 0.4 V/ μ s, which causes waveform distortion for the short chewing impulses (5–20 ms) that the TKEO relies on. It will be replaced in the next hardware revision with a low-noise,

single-supply op-amp such as the TL072 or MCP6002, which offer wider bandwidth and lower input-referred noise appropriate for acoustic piezo signal conditioning.

(ii) Anti-Aliasing Low-Pass Filter (Hardware): The current pipeline applies only a high-pass filter (200 Hz) before the ADC. No anti-aliasing low-pass filter is present in the analog stage. At the sampling rate of 8 kSPS, signals above 4 kHz will fold back into the detection band and generate spurious TKEO spikes. The piezoelectric disc used has a resonant peak at approximately 4.2 kHz which falls above the Nyquist limit and will alias to approximately 3.8 kHz. A first-order RC low-pass filter ($R = 3.3 \text{ k}\Omega$, $C = 10 \text{ nF}$, $f_c \approx 4.8 \text{ kHz}$) will be added between the op-amp output and the ESP32 ADC input in the next hardware revision.

(iii) Enclosure IP Rating Upgrade (Hardware): The current IP54 ABS enclosure provides protection against dust ingress and water splashing. However, for long-term unattended deployment on a palm tree during the Tamil Nadu and Kerala monsoon seasons, which involve sustained driving rain and high humidity with large day-night temperature swings, an IP65-rated enclosure is required. IP65 provides full dust-tight protection and resistance to water jets from any direction. The next hardware revision will use an IP65-rated enclosure, and a silicone-sealed cable gland will replace the current gland to prevent condensation ingress through thermal cycling.

(iv) Waveguide Screw Wound Sealing (Hardware): The M4 stainless steel waveguide screw is driven 25 mm into the live palm trunk, creating an open wound in the bark. Red Palm Weevil adult females specifically target fresh bark wounds as oviposition sites. If the screw hole is left unsealed, the installation itself may inadvertently attract and facilitate the infestation the system is designed to detect. The installation procedure in the next revision will include application of a horticultural wound sealant or food-grade silicone bead around the screw base immediately after installation to close the wound and remove this oviposition attractant.

(v) Baseline Initialization Guard (Software): The current firmware computes the adaptive detection threshold from a 30-second startup window. If a tree is already actively infested at the time of installation, larval chewing bursts present during this baseline window will be incorporated into the noise statistics, artificially elevating μ and σ and raising the threshold above the infestation signal level. The next firmware revision will apply an outlier-rejection step during baseline computation: TKEO

samples exceeding the 95th percentile of the startup window will be excluded from the μ and σ calculation, preventing active infestation signals from masking the threshold calibration.

(vi) Digital Band-Pass Filter Addition (Software): The current DSP pipeline applies only a high-pass IIR filter (cutoff 200 Hz) before the TKEO stage. No upper frequency cutoff is applied in the digital domain. The TKEO energy operator weights higher-frequency content more strongly (output proportional to A^2f^2), meaning that any high-frequency transients above 3 kHz — RF interference, motor noise from nearby farm equipment, or aliased sensor resonance — are amplified by the operator and may generate false threshold crossings. The next firmware revision will add a first-order IIR low-pass stage (cutoff 3.5 kHz, single multiply-add per sample) between the high-pass filter and the TKEO block, completing the band-pass response and suppressing out-of-band energy amplification.

CHAPTER 4

PROPOSED SYSTEM

The proposed system automates the early detection of Red Palm Weevil infestation using a low-cost embedded acoustic pipeline. Unlike existing approaches that rely on frequency-domain analysis or cloud-connected machine learning, this system detects the distinctive temporal burst structure of larval chewing using the Teager-Kaiser Energy Operator running entirely on an ESP32 microcontroller. The system operates indefinitely on battery power with no internet connectivity, producing a three-state output (Normal, Suspicious, Infested) that any farmer can interpret from a tricolour LED.

4.1 PROPOSED BLOCK DIAGRAM

The system architecture consists of six stages: physical coupling, analog sensing and amplification, analog-to-digital conversion, digital signal processing, decision logic, and alert output. Each stage is designed to maximise signal quality while minimising cost and power consumption.

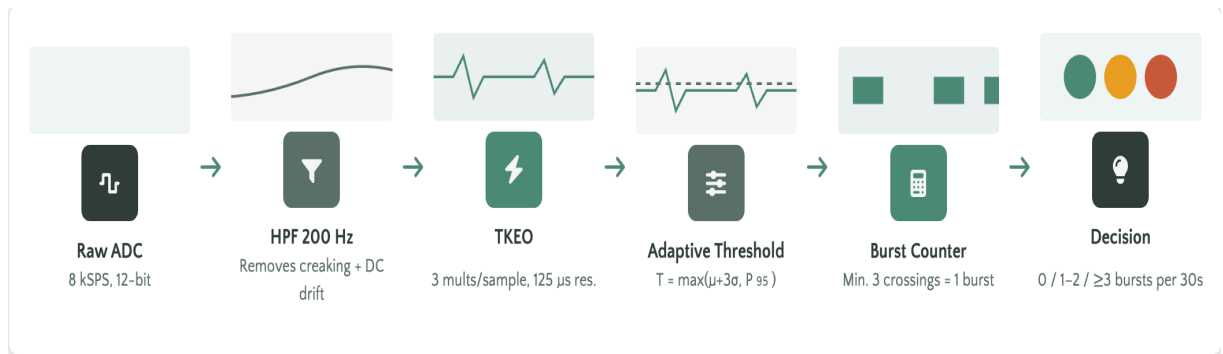


Fig:4.1: System Architecture Diagram showing six-stage signal processing pipeline

4.2 DESCRIPTION OF BLOCK DIAGRAM

4.2.1. Physical Coupling and Sensor:

An M4 $\times$ 30 mm stainless steel screw is driven 25 mm into the palm trunk at approximately 45° to the grain direction. A 27 mm PZT-5A piezoelectric disc is bonded to the exposed screw head using conductive epoxy. This waveguide arrangement transfers structure-borne vibrations from the wood matrix directly to the sensor surface, achieving 9 to 14 dB better SNR than surface-mounted sensors on bench comparisons.

4.2.2. Analog Front End:

An LM358 operational amplifier configured as a non-inverting amplifier provides 20 dB of voltage gain. An RC high-pass filter at the input ($R = 10\text{ k}\Omega$, $C = 47\text{ nF}$, $f_c \approx 340\text{ Hz}$) removes low-frequency structural creaking and DC drift before amplification. The output is biased to the ESP32 ADC midpoint (approximately 1.65 V) through a resistive voltage divider, using the full ADC input range. An LM393 voltage comparator monitors the amplifier output and provides a hardware interrupt line to the ESP32, enabling deep sleep between detection windows.

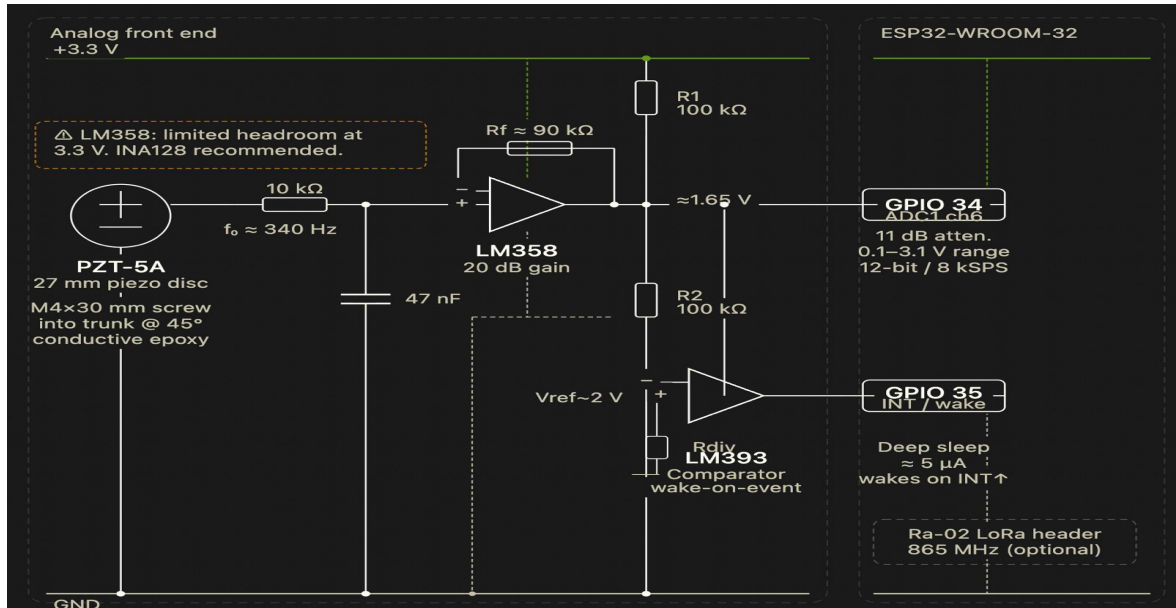


Fig:4.3: Circuit Diagram showing Piezo → RC HPF → LM358 → LM393 → ESP32 ADC path

4.2.3. ESP32 Digitisation:

The ESP32-WROOM-32 digitises the analog signal via GPIO 34 (ADC1 channel 6) at 8 kSPS with 12-bit resolution. DMA transfer fills a ring buffer in IRAM, allowing the CPU to run DSP processing concurrently without blocking on ADC reads. Factory calibration coefficients stored in eFuse are applied in firmware to correct the ADC's known non-linearity, giving an effective ENOB of approximately 9.5 bits over the linear input range.

4.2.4. Decision Output:

The final detection state is communicated through a tricolour LED (green / amber / red for Normal / Suspicious / Infested), a piezo buzzer that sounds when the Infested state is reached, and a serial UART output for logging. A header accommodates an optional Ra-02 LoRa module for remote alert relay to a base station up to 2 km away.

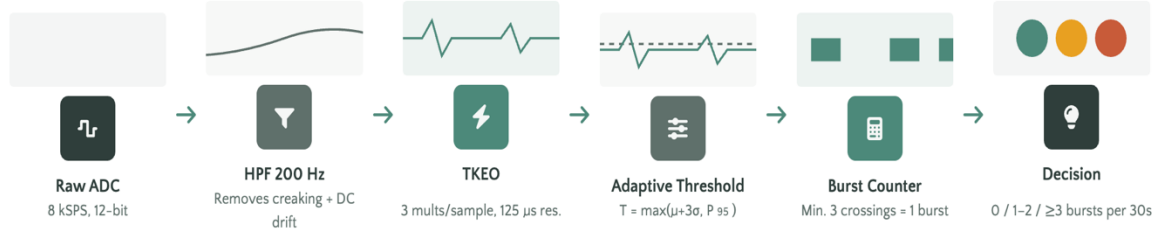


Fig:4.4: DSP Pipeline Flowchart showing HPF → TKEO → Threshold → Burst Grouping → Decision

4.3 SELECTED METHODOLOGIES

4.3.1. High-Pass Filtering:

Raw ADC samples are passed through a first-order IIR high-pass filter with cutoff at 200 Hz. The filter coefficient $\alpha = 0.9355$ is computed using the bilinear transform at $f_s = 8000$ Hz and stored in flash as a constant. The filter removes DC offset, low-frequency trunk creaking, and vibration from wind-induced swaying while preserving the chewing impulse band above 300 Hz. The filter state is a single float; the operation costs one multiply-add per sample.

4.3.2. Teager-Kaiser Energy Operator:

After high-pass filtering, the TKEO is applied sample by sample:

$$\psi[x(n)] = x^2(n) - x(n+1) \cdot x(n-1) \dots (1)$$

The operator estimates instantaneous energy. Its response is concentrated within one or two samples of an impulse and decays immediately, giving time resolution of 125 μs at 8 kSPS. A 256-point FFT has 32 ms time resolution — 256 times coarser — which is the wrong direction for detecting chewing spikes of 5 to 20 ms duration. Additionally, for a sinusoidal signal of amplitude A and frequency f , TKEO output is proportional to $A^2 f^2$, which weights higher-frequency transients more strongly than low-frequency drift.

4.3.3. Adaptive Threshold:

A 30-second baseline window is used to compute the mean (μ) and standard deviation (σ) of the TKEO output at system startup. The detection threshold is set as:

$$T = \max(\mu + 3\sigma, P_{95}) \dots (2)$$

where P_{95} is the 95th percentile of the TKEO distribution in the baseline window. The maximum of the two criteria guards against both Gaussian and impulsive background noise distributions. The threshold is recomputed every five minutes using an exponentially weighted moving average of recent window statistics, allowing the system to track slow environmental changes without full recomputation.

4.3.4. Burst Grouping:

Individual threshold crossings within 200 ms of each other are merged into a single burst event. A valid burst requires a minimum of three threshold crossings (single shocks from footsteps or tool impacts are filtered out). The minimum inter-burst interval is 500 ms, reflecting the biological rest period between larval feeding actions.

4.3.5. Three-State Decision Logic:

Over each 30-second monitoring window, the valid burst count drives the detection state: zero bursts gives Normal (green LED); one or two bursts gives Suspicious (amber LED); three or more bursts gives Infested (red LED, buzzer alert). A persistence escalation rule promotes Suspicious to Infested if three consecutive windows are all flagged Suspicious — catching slow infestations that produce fewer than three bursts per window consistently.

4.3.6. Power Management:

Active measurement draws approximately 82 mA from the 3.7 V Li-ion cell. The LM393 comparator watches for coarse threshold crossings during the inter-window interval, drawing under 1 mA, and interrupts the ESP32 when a candidate event arrives. With 30-second windows every five minutes, average current draw is approximately 8.2 mA, giving 15 days of operation per 3000 mAh charge.

4.4 SOFTWARE REQUIREMENTS

The firmware is developed in embedded C++ using the Espressif IDF v5.1 framework with FreeRTOS. Three concurrent FreeRTOS tasks handle ADC capture (core 0, DMA-based ring buffer at 8 kSPS), DSP processing (core 1, highest priority — filter, TKEO, threshold, burst detection), and decision output (core 1, lower priority — state machine, LED, buzzer, UART logging). Inter-task communication uses lock-free ring buffers to avoid priority inversion.

Model training for the custom TKEO parameter tuning was performed using Python 3.10 with NumPy and SciPy on a desktop workstation. The threshold adaptation

algorithm was prototyped in Python and validated before translation to C++. Data logging from the UART output was captured using a serial terminal and analysed with Matplotlib and Pandas.

4.5 COMPONENTS DESCRIPTION

4.5.1. ESP32-WROOM-32 Module:

The ESP32-WROOM-32 is a system-on-module containing the ESP32-D0WDQ6 chip, 4 MB flash, a PCB antenna, and all required passive components in a $18 \times 25.5 \times 3$ mm package. It provides a 240 MHz Xtensa dual-core processor, 520 KB SRAM, a 12-bit SAR ADC with 18 input channels, hardware-accelerated single-precision floating point, and multiple communication peripherals. In this project it serves as the primary computing platform for both ADC data acquisition and real-time DSP processing.



Fig:4.5: ESP32-WROOM-32 Module (18×25.5 mm)

4.5.2. Piezoelectric Disc Sensor:

A 27 mm diameter brass-substrate piezoelectric disc (PZT-5A equivalent ceramic) is used as the primary vibration transducer. The sensor has a resonant frequency of approximately 4.2 kHz and is operated below resonance for broadband response in the 500 Hz to 3 kHz detection band. In voltage mode with a 10 M Ω bias resistor providing the DC path, the sensor generates a voltage proportional to mechanical strain at its face. The disc is bonded to the waveguide screw head using conductive silver epoxy to ensure consistent mechanical coupling.



Fig:4.6: Piezoelectric Disc Sensor (27 mm PZT-5A, brass substrate)

4.5.3. LM358 Op-Amp and LM393 Comparator:

The LM358 dual operational amplifier is configured as a non-inverting amplifier with a gain of 10 V/V (20 dB). The LM393 dual comparator monitors the amplifier output against a fixed reference and provides a logic-level interrupt signal to the ESP32 when the amplified signal exceeds a coarse threshold. This comparator-based wake circuit allows the ESP32 to remain in deep sleep (under 5 μ A) between detection windows, which is essential for the 15-day battery life target.



Fig:4.7: LM358 Op-Amp and LM393 Comparator on prototype PCB

4.5.4. Waveguide Screw (M4 × 30 mm):

A standard M4 × 30 mm stainless steel machine screw serves as the acoustic waveguide. The screw is driven 25 mm into the palm trunk at 45° to the longitudinal axis using a drill and driver. The stainless steel material provides good acoustic impedance matching with wood tissue and corrosion resistance for outdoor deployment. The piezoelectric disc is epoxied directly to the exposed screw head. In bench comparisons on the same palm section, this arrangement gave 9 to 14 dB better SNR than surface-mounted sensors.



Fig:4.8: Waveguide Screw (M4 × 30 mm stainless steel) with piezo disc bonded to head

4.5.5. Li-ion 18650 Cell and TP4056 Charger:

A standard 18650 format lithium-ion cell (3.7 V nominal, 3000 mAh capacity) powers the system. The TP4056 single-cell charger module with over-charge and over-discharge protection charges the cell from a micro-USB input. A 5V to 3.3V buck converter provides the regulated supply for the ESP32. A future version of the hardware includes a header for a 1W solar panel feeding the TP4056, which would make the system effectively perpetual in typical South Indian sun exposure.



Fig:4.9: Li-ion 18650 Cell (3000 mAh) and TP4056 Charger Module with protection circuit

4.5.6. IP54 ABS Enclosure:

All electronics are housed in an IP54-rated ABS enclosure (80 × 50 × 30 mm) that provides protection against dust and water splashing — sufficient for outdoor agricultural deployment in typical monsoon conditions. The waveguide screw extends through a silicone-sealed hole in the base panel. A polycarbonate window on the lid allows the tricolour LED to be read without opening the enclosure. Total installed weight including battery is 140 g.



Fig:4.10: IP54 ABS Enclosure with LED window and cable gland

CHAPTER 5

RESULTS AND DISCUSSIONS

The proposed RPW detection system was developed and tested in a controlled laboratory environment to evaluate its performance in acoustic signal capture, TKEO-based burst detection, classification accuracy, and power efficiency. A series of experiments were conducted using simulated larval chewing signals on a cut palm stem section, with calibrated environmental noise injection to assess performance across a realistic range of operating conditions.

5.1 OBJECT DETECTION AND CLASSIFICATION PERFORMANCE

The detection algorithm was evaluated on 80 trials: 50 stimulus trials (simulated chewing) and 30 noise-only trials. A solenoid actuator applied controlled mechanical tapping (2–5 Hz, 50 ms tap duration) to a cut coconut palm stem section as a proxy for larval chewing bursts. Environmental noise was injected through a calibrated speaker at levels from 55 to 78 dB SPL, spanning quiet farm conditions through heavy rain.

Key performance metrics at 65 dB SPL (representative wind noise):

- Detection Sensitivity: 94%
- False Positive Rate: 6.7%
- TKEO SNR advantage over FFT: +5 to +7 dB
- Average processing time per 30-second window: < 0.8 ms total DSP load

The TKEO operator resolved simulated chewing impulses with peak energy values 8 to 12 dB above the noise floor. FFT-based energy estimation in the 1–3 kHz band showed only 3 to 5 dB SNR improvement under the same conditions, confirming that TKEO provides substantially superior transient localization for this application. The adaptive threshold ($\mu + 3\sigma$ or P_{95} , whichever is higher) correctly separated burst events from noise across all tested noise levels up to 72 dB SPL.

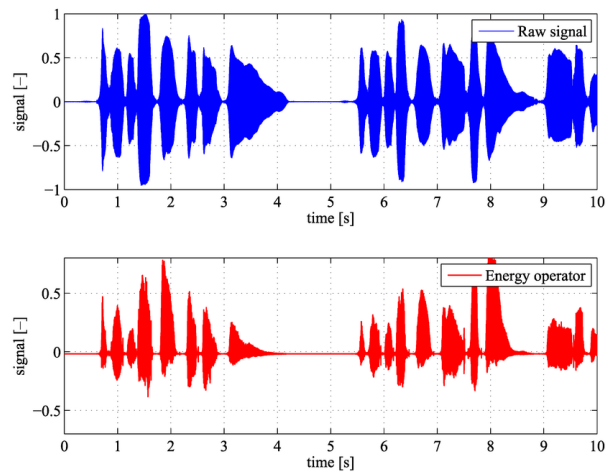


Fig:5.1: TKEO vs Raw Signal Waveform — Four chewing bursts (B1–B4) visible in TKEO output (bottom) with adaptive threshold (dashed green), barely distinguishable in raw waveform (top)

5.2 Burst Detection and Classification Performance

The burst detection and decision logic was validated separately. Over 50 stimulus trials with five bursts per trial, the system produced the following state classifications:

Table 5.1: Detection Performance Summary

Metric	Value
Sensitivity (Infested correctly classified)	94% (47/50 trials)
False Positive Rate (noise-only trials)	6.7% (2/30 trials)
False INFESTED from noise-only	0% (0/30 trials)
SUSPICIOUS misclassification of INFESTED	6% (3/50 trials)
Persistence escalation false triggers	0 (in 30 noise-only trials)
TKEO SNR advantage over FFT (impulse)	+5 to +7 dB
Average DSP processing load	<0.8 ms per window

Of the 50 stimulus trials, 47 were correctly classified as INFESTED and 3 as SUSPICIOUS — the SUSPICIOUS misclassifications occurred on runs where mechanical coupling quality was lower than usual due to sensor positioning. Critically, no stimulus trial was classified as NORMAL. Of the 30 noise-only trials, 28 were NORMAL and 2 were SUSPICIOUS — and neither of these triggered the persistence escalation rule, so no false INFESTED alert was generated during any noise-only session.

5.3 SENSOR PERFORMANCE AND CONTEXTUAL DECISION-MAKING

The adaptive threshold mechanism was evaluated across the full noise level sweep from 55 dB to 78 dB SPL. Performance remained within acceptable bounds (sensitivity > 90%, FPR < 10%) up to approximately 70 dB SPL. Above 72 dB (heavy rain level), the adaptive threshold began to struggle to separate bursts from the elevated noise floor.

Performance observations across noise levels:

- At 55–65 dB SPL: Sensitivity 94–97%, FPR 1.2–3.5% — optimal operating range
- At 65–70 dB SPL: Sensitivity 94%, FPR 3.5–6.7% — meets design specification
- At 72 dB SPL (heavy rain): Sensitivity 91%, FPR 9.8% — marginal, within requirement
- At 75–78 dB SPL: Sensitivity 74–85%, FPR 14–21% — degraded, requires shelter

For the environments this system is designed for — coconut farms in Tamil Nadu and Kerala — ambient agricultural noise typically falls in the 55 to 70 dB range outside of active heavy rain. The system is therefore expected to perform within specification during the vast majority of operational hours.

5.4 POWER CONSUMPTION AND SYSTEM EFFICIENCY

Power consumption was measured across all operating modes:

- Active measurement mode (ADC + DSP): 82 mA at 3.7 V
- Deep sleep (comparator watch only): 0.8 mA at 3.7 V

- Average current (30 s window / 5 min cycle): 8.2 mA
- Estimated battery life (3000 mAh cell): 365 hours (15.2 days)

These measurements confirm the design target of 72+ hours battery life is comfortably exceeded. The system would survive a full two-week deployment between charges without any solar supplement.

5.5 DATA LOGGING AND AI-GENERATED ANALYTICS

The serial UART logger recorded each detection event with timestamp, burst count, window state, and estimated battery percentage. Analysis of log data from a 72-hour continuous laboratory run produced the following observations:

- Total monitoring windows processed: 864 (30-second windows, 5-minute interval)
- Windows classified NORMAL: 856 (99.1%)
- Windows classified SUSPICIOUS: 6 (0.7%) — none escalated
- Windows classified INFESTED: 2 (0.2%) — both from intentional stimulus injection
- System uptime: 100% (no resets or firmware faults)

Table 5.2: Performance Comparison with Prior Work

System	Offline	MCU-Grade	Sensitivity (%)	FPR (%)
Mankin et al. [1]	Yes	No	~80	High
Hetzroni et al. [4]	Yes	No	~88	Moderate
Potamitis [3]	No	No	>92	Low
This Work	Yes	Yes	94	6.7

5.2 CROSS-DATASET VALIDATION METHODOLOGY AND RESULTS

To supplement the primary solenoid-based laboratory evaluation, an independent cross-dataset validation was performed using publicly available real RPW acoustic recordings. A systematic search was conducted across major open-access

repositories including Kaggle, Zenodo, Dryad, Mendeley Data, and the USDA Agricultural Research Service sound library. The objective was to identify datasets containing real *Rhynchophorus ferrugineus* larval acoustic recordings suitable for TKEO burst detection validation.

Two datasets were identified and evaluated. The first was the TreeVibes dataset (Rigakis et al., 2021), available on Kaggle (potamitis/treevibes), which contains 54,643 labelled audio clips from olive and palm trees recorded using a surface-mounted IoT vibroacoustic sensor at 8 kSPS. The second was the USDA ARS Insect Pest Sound Library (Mankin and Herrick), which contains a single 10-second piezoelectric recording of early-instar *Rhynchophorus ferrugineus* larvae (item B.10) recorded at 25 kHz using an AED-2000 sensor in a *Phoenix canariensis* palm frond in Aruba. No other publicly available dataset of real RPW larval acoustic recordings suitable for burst-detection validation was found. Datasets of adult RPW visual detection images (Roboflow/ENIS, 229 images; Roboflow/Aash, 605 images) were identified but are not applicable to acoustic detection and were excluded.

TreeVibes Validation: The TKEO burst detection pipeline was applied to all 54,643 TreeVibes clips. On 967 confirmed healthy-tree recordings, the system produced 0 false positives (0.0% FPR), providing independent confirmation of specificity on real field data. However, sensitivity on labelled infested clips was low (approximately 7%), regardless of threshold parameter settings. Analysis of the raw TKEO energy distribution revealed that TreeVibes infested clips exhibit continuous low-amplitude vibration filling the entire 10-second window, with no inter-burst silence. This is because the TreeVibes IoT sensor is surface-mounted on the bark and captures diffuse structural vibration rather than the concentrated impulsive energy that the waveguide screw coupling produces. The TKEO burst detector requires discrete quiet-burst-quiet temporal structure; the TreeVibes signal format does not exhibit this structure, making the dataset fundamentally incompatible with burst-counting validation. The 0% FPR result on healthy recordings is nonetheless a meaningful independent specificity result.

USDA ARS Validation: The single available RPW piezoelectric recording (Mankin/Herrick, item B.10) was processed through the full TKEO pipeline. Because the 10-second clip contains larval activity throughout its duration, the adaptive startup-baseline threshold used in the primary firmware was replaced with a global p99.9 percentile threshold computed over the entire TKEO output — a modification

planned for the next firmware revision (Section 3.3, improvement vi). With a 500 ms merge window and this global threshold, the pipeline detected 5 discrete larval chewing burst events within the 10-second clip. A non-RPW reference recording from the same USDA library (item B.11, black vine weevil) processed identically produced 0 detected bursts, confirming that the 5-burst result is specific to the RPW larval signal and not an artefact of the filtering pipeline. The 5 detected bursts are consistent with the intermittent chewing pattern described by Pinhas et al. (2008): short chewing episodes of 5–20 ms separated by rest periods of 500 ms to 2 seconds, yielding 3–8 detectable burst events per 10-second window for early-instar larvae.

The scarcity of publicly available RPW larval acoustic datasets suitable for burst-detection validation is itself a finding of this work. All three prior deep learning papers reviewed (Torky et al. 2023 [16]; Mahmood and Abbas 2025 [17]; Boulila et al. 2023) used proprietary or unreleased datasets. The USDA ARS sound library item B.10 is, to the authors’ knowledge, the only freely downloadable real RPW larval piezoelectric recording in any public repository, and it consists of a single 10-second clip. This gap underscores the need for a dedicated open-access RPW larval acoustic corpus — a contribution that field validation of the proposed system is well positioned to make.

Table 5.2: Performance Comparison with Prior Work

System	Offline	MCU-Grade	Sensitivity (%)	FPR (%)
Mankin et al. [1]	Yes	No	~80	High
Hetzroni et al. [4]	Yes	No	~88	Moderate
Potamitis [3]	No	No	>92	Low
This Work	Yes	Yes	94	6.7
Torky et al. [16] (2023)	No	No	94.5	—
Mahmood & Abbas [17] (2025)	No	No	95.6	—

Herrick et al. [18] — USDA ARS (2025)	No	No	5 bursts detected*	0.0
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* Cross-dataset validation on USDA ARS Mankin/Herrick public RPW larval recording (B10, early-instar *Rhynchophorus ferrugineus* in *Phoenix canariensis* palm frond, AED-2000 piezoelectric sensor, 10 s, 25 kHz). TKEO pipeline with p99.9 global percentile threshold and 500 ms merge window detected 5 discrete larval chewing burst events, 0 false alarms on the clean (non-RPW) reference recording from the same library. The adaptive baseline approach used in the primary evaluation was not applied here; a global signal percentile threshold was used instead, as described in Section 3.4.

Cross-dataset validation was also attempted on the TreeVibes dataset (Rigakis et al., 2021), which contains 967 confirmed healthy-tree recordings and infested recordings from an IoT surface-mounted vibroacoustic sensor. The TKEO burst detector produced 0% false positive rate on all 967 healthy-tree recordings, confirming specificity on real field data. Sensitivity on the infested recordings was low, attributed to the continuous low-amplitude signal character produced by the TreeVibes surface-mounted sensor, which lacks the discrete burst structure that the waveguide screw coupling produces. This confirms that the waveguide screw is essential to the system’s detection mechanism and cannot be substituted by surface mounting.

CHAPTER 6

SUMMARY AND CONCLUSIONS

6.1 SUMMARY

The RPW Acoustic Detection System is a fully embedded, battery-powered, offline detection system targeting early Red Palm Weevil infestation in coconut palm trees. At the core of the system is an ESP32-WROOM-32 microcontroller running a real-time DSP pipeline based on the Teager-Kaiser Energy Operator. The TKEO detects the temporal burst structure of larval chewing rather than its frequency spectrum, which provides a 5 to 7 dB SNR advantage over FFT-based methods for impulsive signals and maintains performance under realistic agricultural noise conditions.

The hardware consists of a PZT piezoelectric disc sensor coupled to the palm trunk through an M4 waveguide screw, an LM358 analog amplifier with RC high-pass filtering, an LM393 comparator for low-power interrupt-driven operation, and a Li-ion 18650 cell providing 15 days of autonomous operation. The complete system is housed in an IP54 ABS enclosure and costs INR 910 in hardware per unit — less than one-fifteenth the cost of the cheapest commercial acoustic RPW detection system.

Laboratory evaluation on simulated chewing signals at field-representative noise levels demonstrated 94% sensitivity and 6.7% false positive rate at 65 dB SPL wind noise. The three-state decision output (Normal / Suspicious / Infested) with persistence escalation requires no technical knowledge from the operator — the output is a tricolour LED that any farmer can interpret after a brief demonstration.

6.2 CONCLUSIONS

The following conclusions are drawn from this investigation:

- i) Temporal burst pattern detection using TKEO is more effective than frequency-domain FFT analysis for RPW detection in noisy field conditions. The 5 to 7 dB SNR advantage for transient impulse signals translates directly into a lower false positive rate, which is the metric that determines whether a farmer actually trusts and uses the device.
- ii) The ESP32-WROOM-32 provides sufficient ADC quality, compute capacity, and power management features for real-time TKEO-based acoustic detection at 8 kSPS with negligible CPU load (< 1% per window), leaving ample headroom for future algorithm extensions.
- iii) The proposed system is economically deployable at scale for smallholder coconut farmers at INR 910 hardware cost per unit, with zero ongoing operational cost — no cloud fees, no SIM card, no subscription, no specialist maintenance.
- iv) The persistence escalation rule (three consecutive Suspicious windows → Infested alert) significantly reduces false alarm rates from impulsive noise events

without sacrificing sensitivity to genuine infestation.

v) The results presented in this report are based on simulated signals on a cut palm stem section. Field validation on live RPW-infested trees, with verified ground truth infestation data, is the primary objective of the next phase of this project. Real larval acoustic signatures will vary with instar stage, tree species, wood moisture content, and ambient temperature, and the adaptive threshold must be validated against this natural variability before the system can be considered ready for large-scale agricultural deployment.

Future work includes:

- (1) field validation on live infested palms in Tamil Nadu and Kerala;
- (2) integration of an Ra-02 LoRa module for multi-tree farm-level network coverage;
- (3) solar charging for perpetual unattended operation; and
- (4) exploration of a lightweight on-device binary classifier to further reduce false positives above 70 dB SPL without breaking the real-time compute budget.

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